Master Thesis Proposal
Rewiring the market clock in Spanish electricity markets*

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1 Introduction and literature review

In line with European Commission Regulation (EU) 2017/2195 mandating a 15-minute imbalance settlement period (ISP15) (European Commission, 2017), the **Iberian wholesale electricity market**[1] **transitioned in 2025 from hourly** (60-minute) **to quarter-hourly** (15-minute) **trading intervals**, or Market Time Unit (MTU). Prior to March 2025, both the day-ahead (DA) and intraday (ID) markets cleared hourly products (DA60/ID60), while the ISP15 was introduced in december 2024. In March, intraday trading moved to 15-minute products while the day-ahead market remained hourly (DA60/ID15), and in October the reform was completed with the adoption of 15-minute products in the day-ahead market as well (DA15/ID15). The reform aims to improve the alignment of market schedules with short-run fundamentals, facilitate the integration of variable renewable generation, and reduce the volume of energy balanced close to real time (European Parliament and Council of the European Union, 2019; International Energy Agency, 2025).

The purpose of this thesis is the following:

Purpose of the master thesis

This thesis will empirically assess the impact of the MTU reform and will extend the theoretical framework in my research proposal to better capture firms' strategic behaviour and the system operator's adjustment costs. The thesis will (potentially) have three building blocks:

- **Theoretical model:** extend my previous framework to better capture strategic behaviour and include an explicit welfare analysis (absent from my previous proposal).
- **Reduced-form analysis:** implement a reduced-form evaluation of the reform.
- **Structural modelling:** quantify, from an economic perspective, the price and welfare impacts of counterfactual market designs, in particular the transition DA60/ID60 → DA15/ID60 → DA15/ID15.

*This is an extension of my Energy economics research proposal, focusing on empirical analysis and structural modeling.

[1] Check appendix A for a brief description.

1.1 Literature

The existing related literature is not very vast, and the relevant papers are frequently written in a more engineering style, with limited economic reasoning. An exception is Just and Weber (2015), who propose a simple model to illustrate firms' incentives to manipulate balancing mechanisms prior to ISP15. Similarly, Eicke et al. (2021) reframes balancing as a market equilibrium (with the imbalance price and the system imbalance jointly determined), uses instruments to address endogeneity, and documents feedback from imbalance-price shocks into subsequent intraday prices.

From an empirical perspective, Koch and Maskos (2020) shows that intraday trades respond to information about the system imbalance and can reduce balancing needs in the aggregate, making the MTU15 reform more desirable if market depth is not altered. Along the same lines, Koch and Hirth (2019) finds for Germany that, even with a dramatic expansion of highly volatile generation such as wind and solar, balancing reserves[2] were mainly reduced due to the shift to MTU15, providing strong evidence that market participants respond efficiently to imbalance charges. Moreover, Märkle-Huß et al. (2018) use a Bayesian time series approach to study the change from MTU60 to MTU15 in the European Power Exchange (EPEX) in 2011 and 2014, finding significant price reductions.

Finally, other papers show that adding intraday auctions can improve liquidity/depth and reduce volatility compared to purely continuous intraday trading, and frequent-auction counterfactuals suggest that auctions may lower liquidity costs and deliver less noisy price signals than pure continuous trading (Graf et al., 2024; Neuhoff et al., 2016).

2 Data

There is **abundant publicly available data** to evaluate the reform. In particular, we have access to OMIE data on day-ahead and intraday prices and quantities for the last six rolling years, as well as additional variables of interest (e.g. accepted bid curves, intraday auction outcomes by session, and continuous intraday transactions).

Critically, we can utilize the **disaggregated unit-level bid data** (*Detalle de las ofertas*), with a 3 months lag (currently we have until November 2025). This dataset provides the precise price-quantity pairs submitted by every market participant for each hour of the day. By linking these hourly blocks to unique unit identifiers, we can observe the strategic supply and demand curves[3] of individual power plants and retailers. While these identifiers are anonymized in the short term to preserve market competition, they allow for a granular analysis of how specific technologies and firms adjusted their bidding behavior in response to the reform.

Appendix B provides the list of all the available OMIE data sources.

For controls and mechanisms, weather data can be obtained from AEMET. Data on balancing outcomes and system-operation costs are available from REE via ESIOS, together with other relevant system characteristics such as demand, storage, generation by technology, and interconnection flows. Data collection can be automated using the ESIOS API.

Currently, I am working on a **complete automatization of the data collection to include updated data**. This is particularly important for the dissagregated data, since there is a 3 month lag and therefore we only currently see less than 2 months after the DA15 reform of october.

As to February 23th, I have already the automated code for DA prices and all the collection architecture.

[2] Part of the system operator's balancing mechanisms.

[3] Positive and negative quantities.

3 Roadmap

Here I propose strategies to be discussed in my thesis. Not all of them have to be covered in the same way, and the idea is to focus more on the empirical part.

3.1 Economic theory

Summary of my research proposal model In my research proposal I build on the sequential-market framework of Ito and Reguant (2016) to study how the MTU reform changes price wedges, within-hour dispersion, and balancing needs. I model two stages, day-ahead (DA) and intraday (ID), and represent MTU60 versus MTU15 by splitting the hour into two subperiod products. A residual monopolist faces linear residual demand in each subperiod, while intraday trading reallocates quantities around the DA commitment with lower elasticity close to delivery ($b_{1t} > b_{2t}$), capturing thinness. I also allow thinness to differ across subperiods (e.g. $b_{21} \neq b_{22}$), which becomes central in the transitional DA60/ID15 regime.

I find that the reform operates through two distinct channels. First, product alignment tightens the no-arbitrage restrictions: with DA60/ID60 competitive arbitrage yields $p_1 = p_2$, with DA60/ID15 it implies only block parity $p_1 = (p_{21} + p_{22})/2$ (so dispersion arises whenever $b_{21} \neq b_{22}$), and with DA15/ID15 it implies product-by-product parity $p_{1t} = p_{2t}$. Under strategic arbitrage, wedges persist in all regimes because the arbitrageur internalizes price impact. Second, DA granularity drives balancing: with DA60, within-hour fundamentals μ_t necessarily spill into balancing ($\Delta_t = \mu_t + \varepsilon_t$), while with DA15 they are internalized in schedules ($\Delta_t = \varepsilon_t$). Overall, I conclude that the **transitional DA60/ID15 step can increase within-hour intraday dispersion without reducing balancing needs**, whereas the **DA15/ID15 step targets efficiency directly by removing the systematic within-hour component from balancing**, even if market-power wedges remain in thin near-delivery products.

Ideas to (potentially) extend

- Include a distinction between sophisticated and non-sophisticated agents into my Ito and Reguant (2016) model extension to reflect winners and losers in this new reform and how its proportion affects final prices (maybe due to higher balancing penalties).
- Include a complete welfare analysis and extend the model to the rest of strategic cases in Ito and Reguant (2016).
- Analyze how MTU changes outcomes in continuous markets versus frequent batch auctions (Graf et al., 2024), as finer MTUs can amplify microstructure noise in thin near-delivery order books (not sure about this).

3.2 Reduced form analysis

There are many outcomes of interest to study the effect of the transition, so I will cover the most relevant:

3.2.1 Outcomes of interest

Arbitrage Price Wedge: It measures the price difference between sequential market stages for the same delivery period p, which could refer to a pair of hour and quarter (h, q) in case of MTU15 or just an h for MTU60. In DA60-ID60, the arbitrage price wedge would be:

$$W_{d,h}^{DA/IDA} = P_{d,h}^{DA} - \bar{P}_{d,h}^{ID}$$

where $\bar{P}_{d,h}^{ID}$ is a volume weighted average price (VWAP) of the intraday auctions:

$$\bar{P}_{d,h}^{ID} = \sum_{j \in \text{IDauctions}} \frac{\text{volume}_{d,h,j}}{\text{totalvolume}_{d,h}} P_{d,h}^{ID(j)}$$

This could be easily extended to the wedge between the intraday auctions and the continuous time market, using a similar VWAP for a particular interval or window W relative to the delivery time of interest[4]:

$$\hat{P}_{d,h}^{CIM}(W) = \sum_{l \in \mathcal{L}_{d,h}(W)} p_l \frac{Q_l}{\sum_{\ell \in \mathcal{L}_{d,h}(W)} Q_\ell}$$

where $\mathcal{L}_{d,h}(W)$ is the set of trades for the period (d, h) inside the window W, which could be, for example, 2 hours before delivery. The arbitrage price wedge in this case could be measured in the same way with respect to the DA, $W^{DA/CIM}$, or to the IDAs, $W^{IDA/MIC}$.

Note that in an efficient market, the Day-Ahead (DA) price should equal the expected Intraday (ID) prices; a persistent wedge indicates either strategic market power or technical barriers to arbitrage.

In the DA60-ID15 scenario, arbitrage price wedge could be measured using each quarter, $W_{d,h,q}^{DA/IDA} = P_{d,h}^{DA} - \bar{P}_{d,h,q}^{ID}$, or using the average over the quarters for the intraday prices. For the DA15-ID15, we would consider quarter to quarter.

Finally, another to see whether MTU15 is making markets more efficient is to see how $\hat{P}^{CIM}$ change when considering different windows W: if in the MTU15 the contracts in an early window have similar prices than the contracts in a late window close to delivery after controlling for covariates, we could infer that MTU15 is making the market more efficient.

Liquidity and thinness We have data on the number of trades and their volume for the continuous market, so we can easily see whether the total hourly volume increased after MTU15 or it remained constant. *A priori*, larger volume would be beneficial for the efficiency of the system, since more real time information would be included in the trades.

Schedule revision intensity ($\Delta Q_{d,\tau}$): This measures the physical gap between the initial market plan and the final delivery schedule. In MTU60, generators cannot schedule their sub-hourly variations, forcing the System Operator (REE) to activate costly balancing reserves to maintain frequency. Therefore, under MTU15 we would expect the market outcomes to be very similar to the validated outcomes of REE.

[4]Note that the market is continuous in the sense that it can be negotiated in continuous time, but the MTU is still fixed.

3.2.2 Econometric strategy

The main issue is that **we do not have a clear counterfactual**: most countries in Europe are already in the MTU15 system and they are not directly comparable to Spain, in particular due to differences in renewable penetration (solar and wind).

For this reason, the existing literature typically compares prices before and after the reform, controlling for a large set of covariates, in the hope of isolating the effect of the policy.

Another novel strategy, used in Märkle-Huß et al. (2018) for the European Power Exchange (EPEX) MTU reform (2011 and 2014), is to apply a **Bayesian structural time series approach**, as proposed by Brodersen et al. (2015). The idea is to construct a data-driven counterfactual of the outcome variable using only pre-intervention information.

More precisely, the method estimates a **structural state-space model in the pre-reform period**, including components for local trend, seasonality, and a set of contemporaneous covariates (e.g. renewable generation, demand, fuel prices). A spike-and-slab prior is used for variable selection, allowing the model to build a synthetic control from the available predictors while avoiding overfitting.

Using the posterior distribution of the model parameters, the framework generates the posterior predictive distribution of the outcome variable in the post-reform period under the assumption that the reform had not taken place. Denoting the observed series by y_t and the counterfactual prediction by $\tilde{y}_t$, the pointwise causal effect is defined as

$$V_t = y_t - \tilde{y}_t.$$

The method provides the full posterior distribution of V_t, as well as the accumulated effect over the post-treatment period, together with credible intervals and posterior probabilities of a non-zero impact.

This approach generalizes difference-in-differences to a time-series setting and allows for flexible dynamics, while explicitly quantifying uncertainty around the estimated reform effect.

As to February 24, I am still focused on data collection, so I have not checked in detail this method.

Choice of pre-reform periods. Selecting the pre-intervention estimation window is not trivial. The most important reforms for **wholesale** market prices are ID15 and DA15; ISP15 affects primarily security and balancing/imbalance-settlement mechanisms. Therefore, for ID15 it is natural to use all available pre-March years, but the appropriate window may still depend on how strongly balancing conditions feed into the outcome of interest (e.g. wedges close to delivery or continuous-market liquidity).

What about the April 28 blackout? The April 28 disturbance is a major shock that can confound the assessment of MTU15. It affects both the **post-ID15** period and the **pre-DA15** period. For ID15, a natural approach is to focus on the post-reform window *before* the blackout (March 19–April 28), potentially excluding additional days with unusually large imbalances (e.g. April 22). For DA15, the evaluation necessarily takes place in a post-blackout environment, so it is useful to separate a blackout-to-reform period (April 28–October 1) from the post-reform period. Moreover, REE increased security requirements and balancing reserves as a precaution, so the impact assessment of DA15 should be interpreted in this tighter balancing context.

3.3 Structural model

A richer alternative would be a **firm-level structural bidding model** (potentially by technology) exploiting the detailed bid data published by OMIE. The *Detalle de las ofertas al mercado diario* files report, for each unit and delivery period, the submitted price–quantity steps and the corresponding accepted quantities, which makes it possible to reconstruct full bid schedules rather than relying solely on market-clearing prices.

This is the most ambitious component of the project. Despite the three-month publication lag, the disaggregated bidding data are sufficiently rich to support structural analysis, including information that allows bids to be linked to technology and plant ownership.

The main advantage of such a structural approach is that it shifts the focus from average price effects to the underlying economic mechanisms. In particular, it would allow us to address the following questions:

- Does the MTU reform change the elasticity of residual demand faced by firms (i.e. market thinness)?
- Are observed price changes driven primarily by fundamentals (renewables, demand, fuel costs), or by changes in strategic bidding behaviour?
- Do firms adjust the curvature or steepness of their bid curves when products become more granular?
- Are flexible technologies (e.g. hydro, CCGT) more responsive to the MTU reform than less flexible technologies?
- Does the reform shift strategic activity between the day-ahead and intraday markets?

Rather than focusing exclusively on average price differences, a structural perspective would therefore allow the impact of MTU15 to be decomposed into changes in market power, market thickness, and intertemporal reallocation of quantities.

A closely related exercise is conducted in Hortaçsu et al. (2017), who study whether more sophisticated firms in the Texas electricity market deviate more from Nash equilibrium bidding, finding that greater sophistication is associated with improved system efficiency. A general reference for this class of models is Gentry et al. (2018), although multi-unit uniform-price auctions are not covered in detail.

The main limitation of this approach is that public OMIE/REE data do not provide information on forward contract positions or unit-level marginal costs. As a result, identification would necessarily rely on additional assumptions or credible cost proxies.

Finally, the April 2025 disturbance can be interpreted as a change in market tightness or balancing conditions. This makes it possible to examine whether bidding behaviour and market thickness respond differently under more stressed system conditions, without requiring a full simulation of the transmission network.

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Appendix

A Spanish electricity market

Spain participates in the Iberian market (MIBEL). The nominated electricity market operator (NEMO) is OMIE (Operador del Mercado Ibérico de Energía), which organises the main short-term energy markets. In these markets, participants (generators, retailers, large consumers and other market agents) submit bids/offers for electricity for each delivery period (market time unit, MTU). Market outcomes determine market prices and a set of scheduled energy programmes (i.e., planned injections and withdrawals) for each MTU.

In particular, short-term trading is organised through the following segments:

- **Day-Ahead Market (DA, SDAC):** the first and largest stage of short-term electricity trading for delivery on the next day. It is a multi-unit, uniform-price (marginal pricing) auction cleared through the Single Day-Ahead Coupling (SDAC). The output is a day-ahead price and a scheduled programme for each delivery period.

- **Intraday Auctions (IDA):** several discrete intraday auction sessions allow participants to update their positions as forecasts and unit availability change. These auctions also follow a multi-unit, uniform-price format and are integrated in the European intraday auction framework (where applicable).

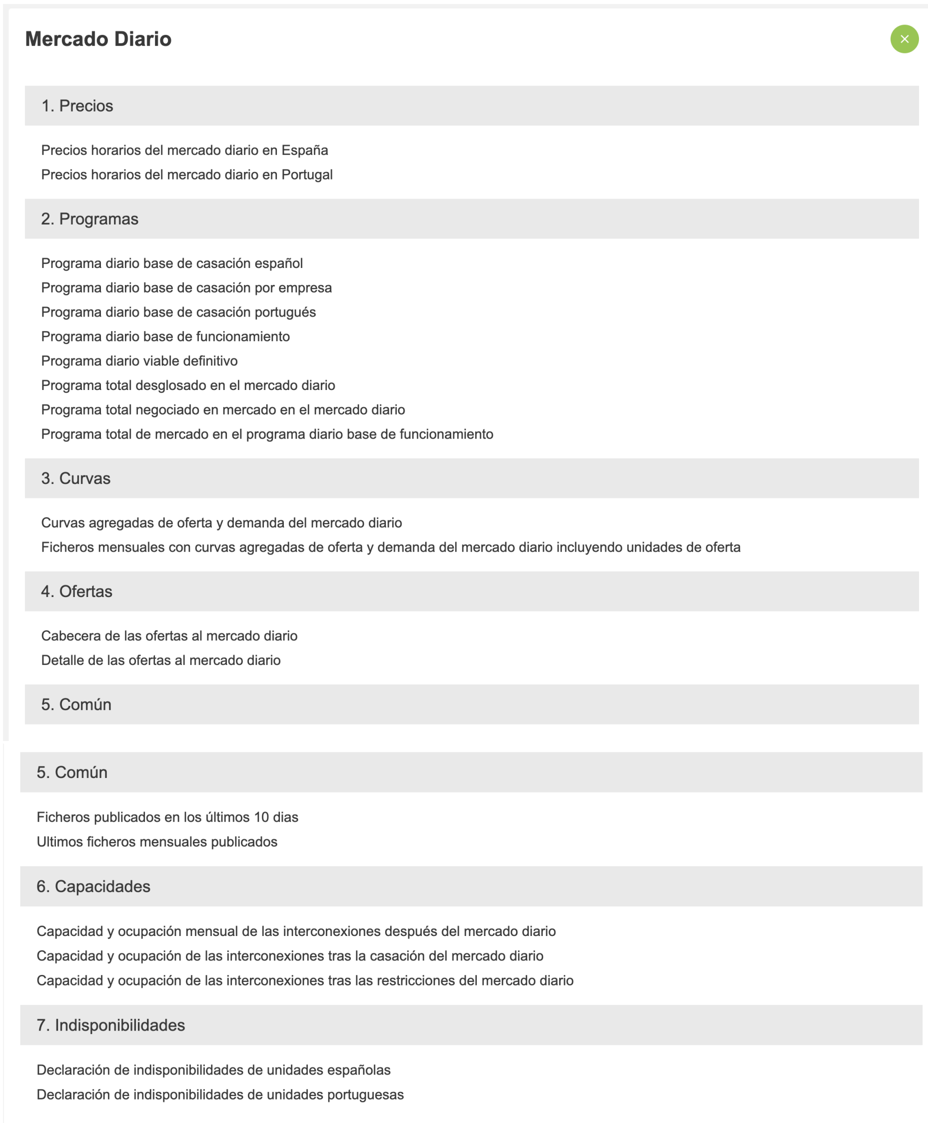
- **Continuous intraday market (SIDC/XBID):** a continuous order-driven market where buy and sell orders are matched continuously. Trades occur at transaction prices (there is no single uniform clearing price), and this segment provides the last opportunity to voluntarily adjust positions close to real time.

Because supply and demand must balance physically at all times, Spain has a transmission system operator (TSO) and system operator, Red Eléctrica (REE). REE is responsible for secure system operation: it performs feasibility/security checks of market-based schedules (network constraints, operational limits, adequacy of reserves), and it may apply additional system services to ensure that the final schedules are physically deliverable.

After intraday trading closes for a given delivery period (i.e., once voluntary trading opportunities are exhausted), any remaining mismatch between an agent's realised metered position and its final scheduled programme becomes an *imbalance*. REE manages these real-time deviations by activating balancing and ancillary services (e.g., automatic and manual frequency restoration and other adjustment services) to maintain system balance and frequency. Deviations are subsequently settled financially in the imbalance settlement mechanism at imbalance prices defined for each imbalance settlement period (aligned with the MTU).

B OMIE data

I include screenshots from the OMIE webpage



Mercado Diario

1. Precios

Precios horarios del mercado diario en España
Precios horarios del mercado diario en Portugal

2. Programas

Programa diario base de casación español
Programa diario base de casación por empresa
Programa diario base de casación portugués
Programa diario base de funcionamiento
Programa diario viable definitivo
Programa total desglosado en el mercado diario
Programa total negociado en mercado en el mercado diario
Programa total de mercado en el programa diario base de funcionamiento

3. Curvas

Curvas agregadas de oferta y demanda del mercado diario
Ficheros mensuales con curvas agregadas de oferta y demanda del mercado diario incluyendo unidades de oferta

4. Ofertas

Cabecera de las ofertas al mercado diario
Detalle de las ofertas al mercado diario

5. Común

5. Común

Ficheros publicados en los últimos 10 dias
Ultimos ficheros mensuales publicados

6. Capacidades

Capacidad y ocupación mensual de las interconexiones después del mercado diario
Capacidad y ocupación de las interconexiones tras la casación del mercado diario
Capacidad y ocupación de las interconexiones tras las restricciones del mercado diario

7. Indisponibilidades

Declaración de indisponibilidades de unidades españolas
Declaración de indisponibilidades de unidades portuguesas

Figure 1: Day Ahead Market data

Mercado Intradiario

1. Precios

Precios del mercado intradiario de subastas en España
Precios del mercado intradiario de subastas en Portugal

2. Programas

Programa horario final
Programa acumulado por unidad de oferta tras cada sesión del mercado intradiario de subastas
Programa incremental de unidades españolas en cada sesión del mercado intradiario de subastas
Programa incremental por empresa en cada sesión del mercado intradiario de subastas
Programa incremental de unidades portuguesas en cada sesión del mercado intradiario de subastas
Programa total desglosado tras el mercado intradiario de subastas
Programa total desglosado por sesion tras el mercado intradiario de subastas
Programa incremental de las unidades de producción y adquisición en cada sesión del mercado intradiario de subastas
Programa total negociado tras el mercado intradiario de subastas
Programa incremental total en cada sesión mercado intradiario de subastas

3. Curvas

Curvas agregadas de oferta y demanda del mercado intradiario de subastas
Ficheros mensuales con curvas agregadas de oferta y demanda del mercado intradiario de subastas incluyendo unidades de oferta

4. Ofertas

Cabecera de las ofertas al mercado intadiario de subastas
Detalle de las ofertas al mercado intadiario de subastas
Limitaciones a las unidades españolas
Limitaciones a las unidades portuguesas

5. Común

Ficheros publicados en los últimos 10 dias
Ultimos ficheros mensuales publicados

6. Capacidades

Capacidad y ocupación de las interconexiones después de los mercados intradiarios

7. Indisponibilidades

Declaración de indisponibilidades de unidades españolas
Declaración de indisponibilidades de unidades portuguesas

8. Otros

Horas anuladas en el programa horario final español por el Operador del Mercado
Horas anuladas en el programa horario final español por el Operador del Sistema
Horas anuladas del programa horario final portugués por el Operador del Mercado
Horas anuladas en el programa horario final portugués por el Operador del Sistema

Figure 2: Intraday auction data

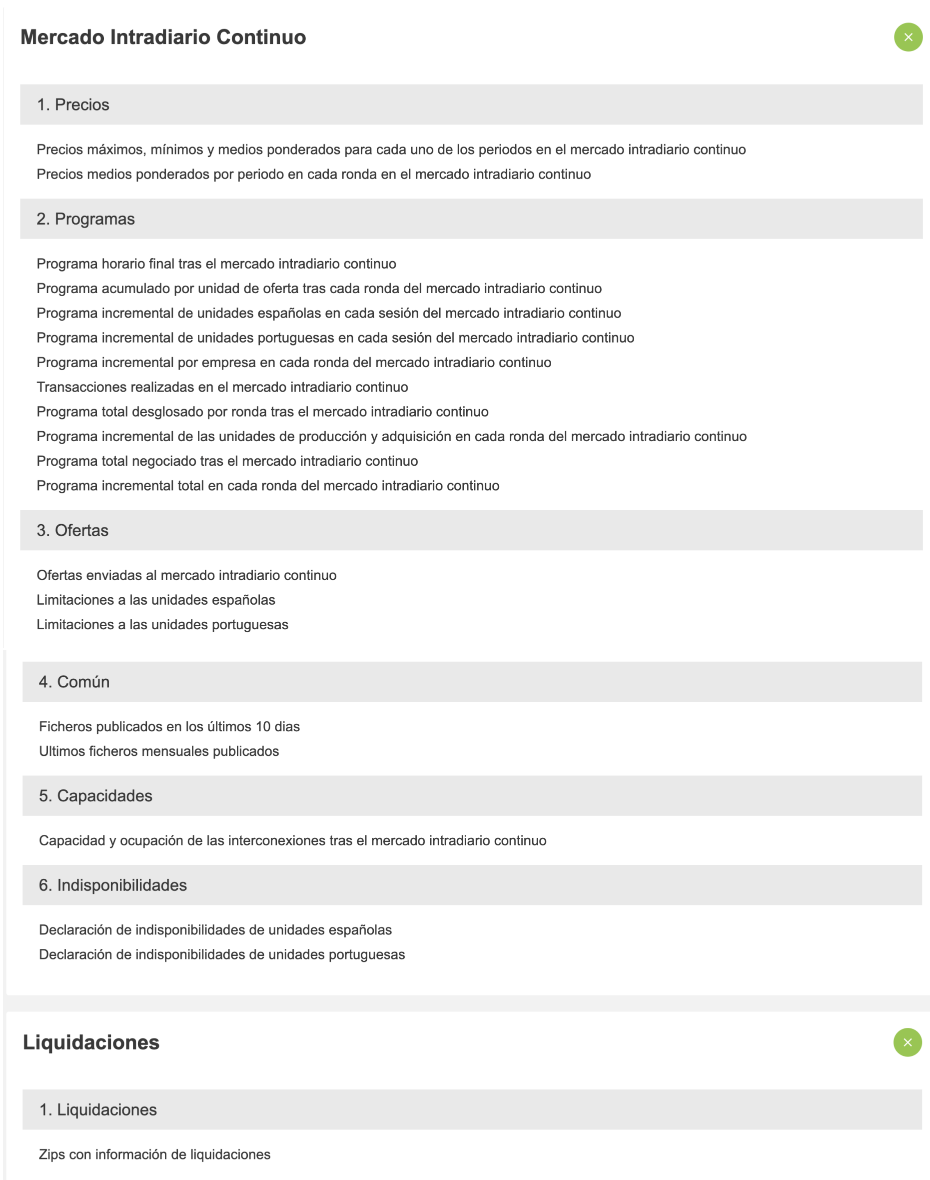


Figure 3: Continuous Market data